

Base Models Look Human To AI Detectors

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Abstract

As AI-generated text enters the real-world at scale, institutions increasingly use commercial AI-text detectors, especially in education and academic-integrity workflows. We report a surprising empirical finding about such systems: when evaluated by GPTZero and Pangram, generated text from base models is often judged overwhelmingly human, whereas text generated by their instruction-tuned counterparts is not. Building on this observation, we propose *Humanization by Iterative Paraphrasing (HIP)*, a detector-agnostic pipeline that minimally fine-tunes a base model into a paraphraser and applies it iteratively. Compared with the baselines we test, HIP yields a stronger trade-off between semantic preservation and detector evasion on commercial detectors. Across Llama-3 and Qwen-3 families, spanning model sizes from 0.6B to 70B, HIP consistently improves detector human-likeness. Our findings suggest that current detectors are tracking artifacts of instruction tuning and local context more than any invariant notion of machine-generated text. This, in turn, calls for detector designs that model these factors more explicitly.

1 Introduction

As large language model (LLM) text becomes commonplace, distinguishing human-written text from machine-generated text has become a practical problem rather than a purely academic one. Commercial LLM-text detection systems such as GPTZero [Adam et al., 2026] and Pangram [Emi and Spero, 2024] have emerged, and they have been deployed in real-world use cases including assignment screening and authorship review [GPTZero, 2026, Pangram, 2026].

At the same time, a growing body of work studies how to evade such detectors by treating them as optimization targets. This includes paraphrasing-based rewriting and, more recently, reinforcement-learning-based methods that optimize directly against detector APIs [David and Gervais, 2025, Ranganath and Ramesh, 2026]. Our work begins one step earlier: *are there models whose outputs commercial detectors already judge to be human-written, without detector-aware optimization?*

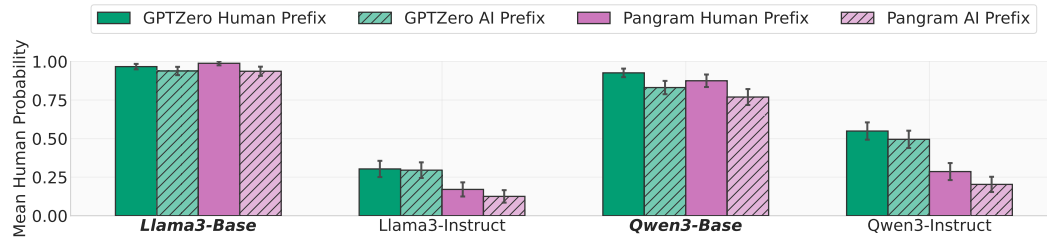


Figure 1: GPTZero and Pangram human-probability scores on text generated by base and instruction-tuned Llama3-8B and Qwen3-8B models when conditioned on human-written or AI-generated prefixes. The error bars show 95% confidence intervals. Across both model families, base-model continuations are judged substantially more human than instruct-model continuations.

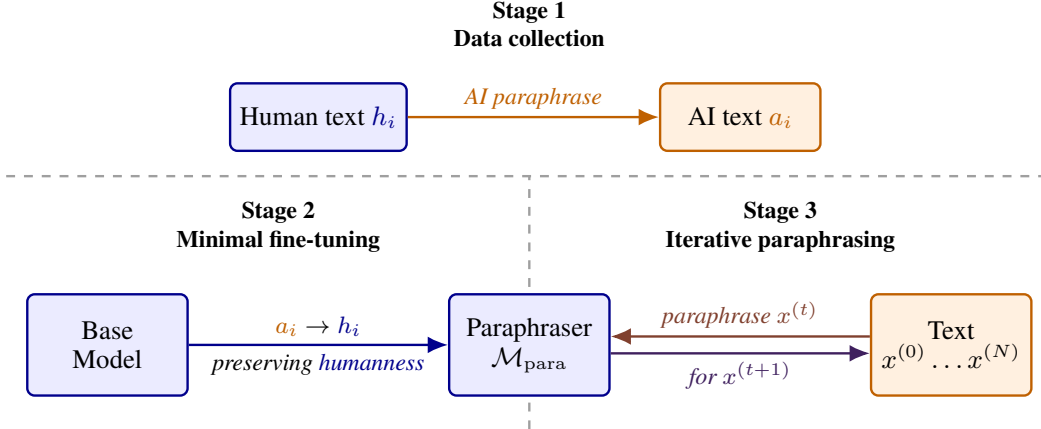


Figure 2: Overview of Humanization by Iterative Paraphrasing (HIP). **Stage 1 (data collection):** an AI paraphraser rewrites each human passage h_i into an AI counterpart a_i , yielding paired data $\mathcal{D} = \{(a_i, h_i)\}$. **Stage 2 (minimal fine-tuning):** a base model is lightly adapted on $a_i \rightarrow h_i$ to obtain a paraphraser $\mathcal{M}_{\text{para}}$ while preserving native continuation behavior as much as possible. **Stage 3 (iterative paraphrasing):** $\mathcal{M}_{\text{para}}$ is applied repeatedly to an input passage, producing $x^{(0)}, \dots, x^{(N)}$ whose outputs become progressively more human-like to current detectors.

The answer is yes. Current commercial detectors judge base-model continuations far more human than instruction-tuned continuations. To show this, we directly evaluate Llama-3-8B and Qwen3-8B under human-written and AI-generated single-sentence prefixes. Figure 1 summarizes the result. For Llama-3-8B with human prefixes, GPTZero and Pangram assign human probabilities of 96.7% and 98.8% to the base model’s continuations, respectively, and 30.3% and 17.1% to the instruct model’s continuations. Similar gaps appear under AI prefixes and on Qwen3-8B.

These measurements suggest two working intuitions about what makes model outputs look human to current detectors. The first is *low distortion*: outputs closer to base-model continuation behavior are judged more human than outputs produced after instruction tuning. The second is *human context*: human prefixes make model continuations look slightly more human than AI prefixes. In other words, conditioning on text already drawn from the human-written distribution can shift subsequent continuations in a more human-looking direction from the perspective of current detectors.

The observations motivate a detector-agnostic rewriting pipeline. We minimally fine-tune a base model into a paraphraser while keeping it close to base-model continuation behavior, thereby preserving *low distortion*. We then apply it iteratively so that the local context is progressively rewritten away from the original AI text and toward *human context*. We call this pipeline *Humanization by Iterative Paraphrasing (HIP)* and illustrate it in Fig. 2. Across Llama and Qwen models of multiple sizes, HIP yields a stronger trade-off between semantic retention and detector evasion on the state-of-the-art commercial detectors we study than the previous approaches we test, including simple prompt-based paraphrasing, supervised paraphrasing baselines [Krishna et al., 2023], Unicode-substitution baselines [Creo and Pudasaini, 2025], and reinforcement-learning-based detector-evasion methods [Ranganath and Ramesh, 2026]. Moreover, unlike much of the academic literature, which evaluates primarily on open-source detectors, we conduct this evaluation on state-of-the-art commercial detectors.

We summarize our contributions as follows.

- We identify a surprising empirical pattern on commercial detectors: base-model continuations are judged substantially more human than instruction-tuned continuations under the same prefix conditions, which motivates two intuitions about what makes model outputs look human to current detectors: *low distortion* and *human context*.
- We introduce *Humanization by Iterative Paraphrasing (HIP)*, a detector-agnostic pipeline that minimally adapts a base model into a paraphraser and applies it iteratively to humanize AI-generated text. Empirically, HIP works across Llama and Qwen model families and a range of model sizes, yielding a stronger semantic-evasion trade-off than the previous approaches we test.

- We point to detector-side research directions, arguing that future systems should pay attention to base-model behavior, post-training distortions, and local context more explicitly.

2 Related Work

AI text detection. As LLMs have advanced, detecting AI-generated text has become an important practical problem. Existing methods include zero-shot or statistical approaches, such as DetectGPT [Mitchell et al., 2023] and Binoculars [Hans et al., 2024], as well as supervised classifiers trained on labeled human and machine text. Commercial detectors such as Pangram [Emi and Spero, 2024] and GPTZero [Adam et al., 2026] report strong cross-domain performance using supervised neural classifiers trained on large corpora of human- and machine-written text. As LLMs are increasingly used as collaborative co-authors rather than sole generators, the boundary between human and machine text is also blurring. Thai et al. [2025] move beyond binary classification by quantifying the extent of AI editing, while MixSet [Zhang et al., 2024] evaluates detectors in subtle revision and mixed-authorship settings. Much of this literature evaluates text produced directly by assistant-style or post-trained models. Our paper instead asks how current detectors behave on unmodified base-model continuations, especially under human-written prefix context.

Behavior shift during post-training. Instruction tuning and RLHF leave statistical fingerprints that can be both characterized and partially reversed. On the characterization side, Casper et al. [2023] list distributional shift as a central concern of post-training, and concrete artifacts have been documented including response length [Singhal et al., 2024] and sycophancy [Sharma et al., 2024]. Movva et al. [2026] use sparse autoencoders to analyze preference datasets, finding that LMArena strongly favors Markdown-style formatting with headings, lists, and bolded text. On the reversibility side, Jindal et al. [2025] document that continual pretraining significantly degrades instruction performance, and Morris [2025] recover a base-like model from the post-trained GPT-OSS-20B via low-rank fine-tuning on pre-training data. Our paper contributes to both strands: we use detector behavior as an empirical lens on post-training shifts, and we find that benign continued exposure to base-style data is sufficient to recover detector human-likeness without any detector-aware optimization.

Adversarial paraphrasing and detector evasion. The deployment of AI text detectors has been accompanied by a growing line of research on how to evade them. Sadasivan et al. [2023] analyze paraphrasing as a fundamental weakness of many detectors, and DAMAGE [Masrour et al., 2025] studies detectors on humanized AI text while proposing a more robust detector. Recent methods include temperature-guided paraphrasing such as TempParaphraser [Huang et al., 2025], supervised rewriting models such as DIPPER [Krishna et al., 2023], orthographic attacks based on homoglyph substitution such as SilverSpeak [Creo and Pudasaini, 2025], style-humanization approaches such as MASH [Gu et al., 2026], and reinforcement-learning-based attacks such as AuthorMist [David and Gervais, 2025] and StealthRL [Ranganath and Ramesh, 2026], which optimize against black-box detector APIs. Beyond the academic literature, commercial AI humanizers are also now marketed explicitly as detector-evasion tools, and recent academic work has begun to study such systems systematically [Masrour et al., 2025]. Our paper studies detector evasion in a different regime: we use minimal adaptation to exploit a human-like behavior already present in base-model generations, evaluate on state-of-the-art commercial detectors rather than only on open or research detectors, and use the observed behavior to point toward new research directions for detectors.

Contextual influence and iterative refinement. The context in which an LLM operates strongly influences its generation distribution, so iterative rewriting has become a natural setting for detector evasion. TH-Bench [Zheng et al., 2025] studies humanization attacks against detectors, while PADBen [Zha et al., 2025] specifically analyzes iterative paraphrasing and benchmarks robustness to paraphrase attacks. Beyond evasion, iterative refinement is also a general capability of modern LLMs. Self-Refine [Madaan et al., 2023] shows that a single model can improve outputs through repeated feedback-and-revision cycles. Our paper connects these strands by asking whether iterative paraphrasing can progressively replace AI-origin context with more human-looking context.

3 Methodology

We have seen in Section 1 that base models, when conditioned on human text, are overwhelmingly detected as human by current detectors. As discussed in Section 1, this phenomenon suggests two